

Activity 8: Chatroom REST API + UI

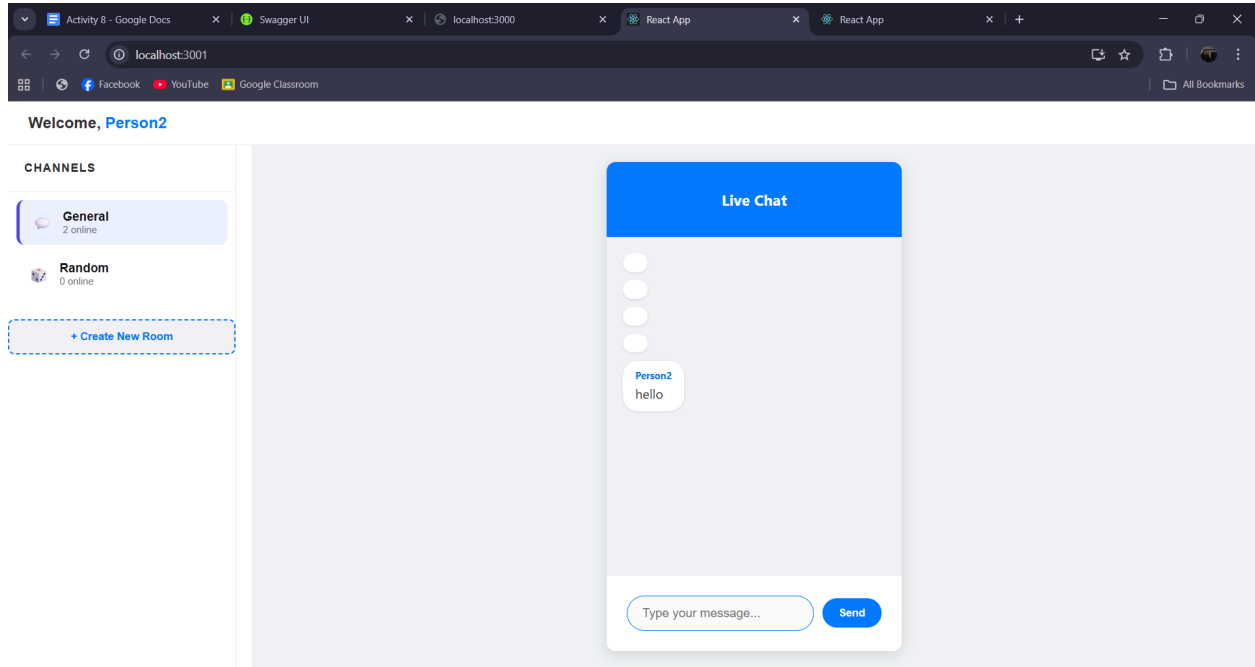
Leader:

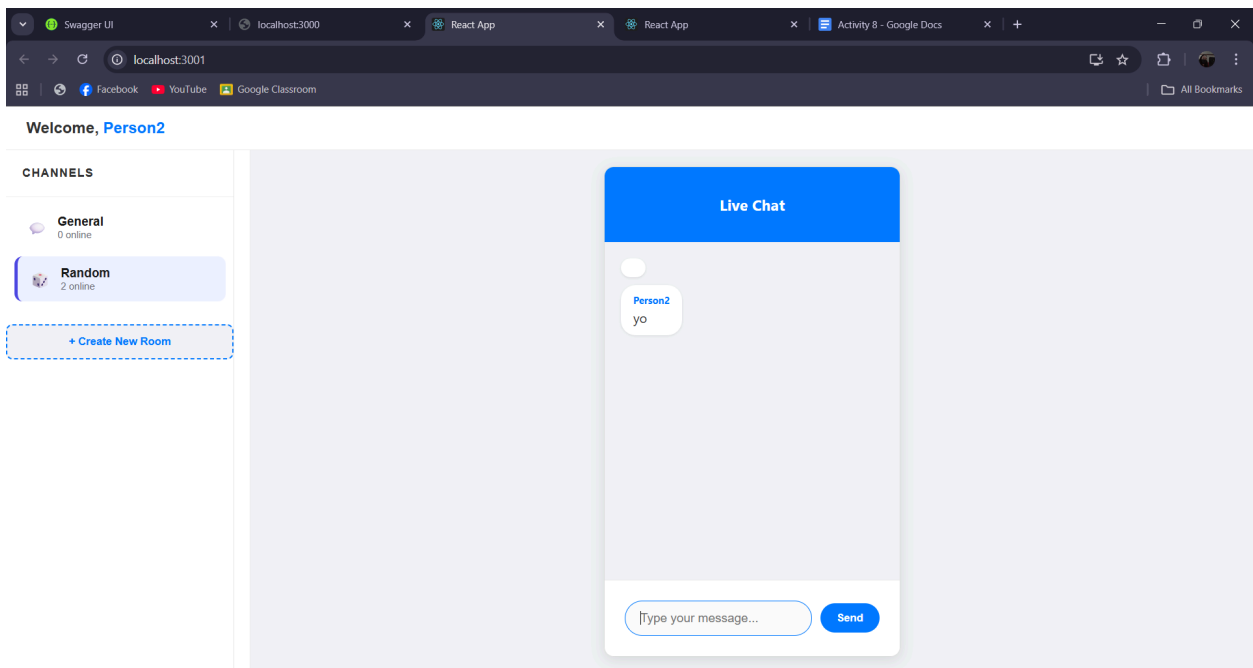
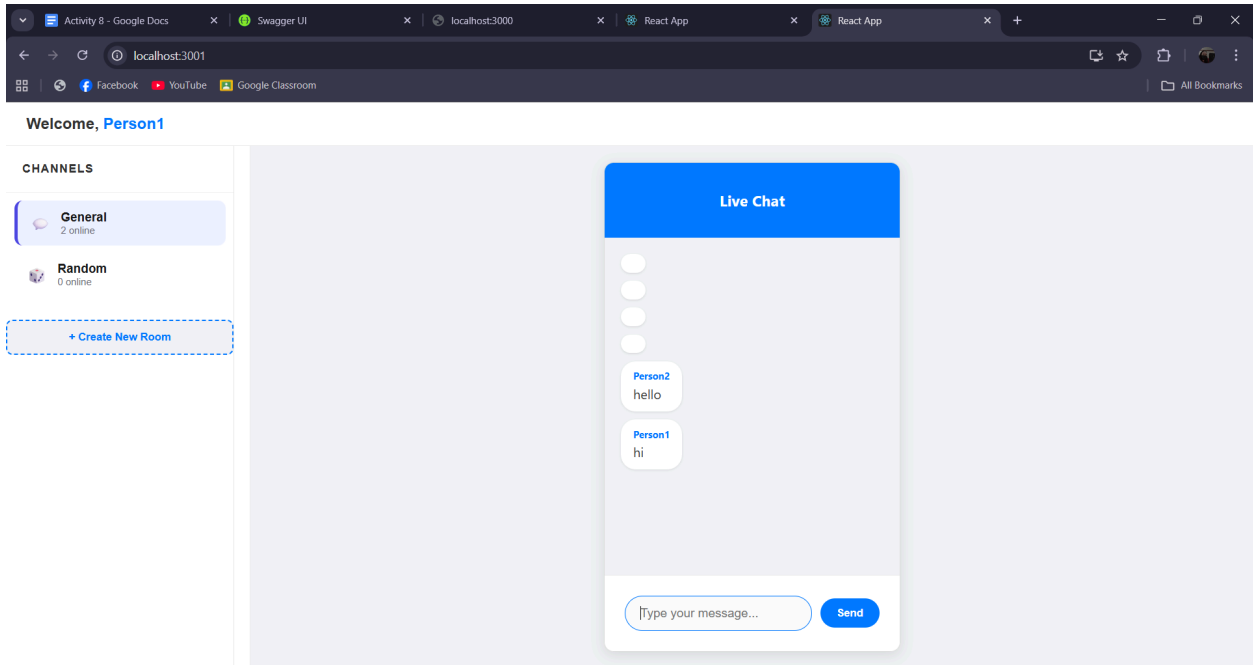
Darryl Jedidiah A. Aranzasu

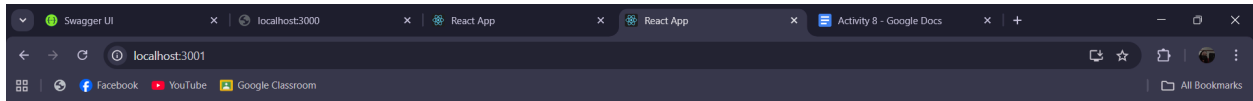
Members:

Czyrell Gwen R. Abelita
Darrylene Kaela A. Aranzasu
Demrose Carla C. Gangan
Andrie B. Moral
Nathaniel L. Relox

Activity 8: Chatroom REST API + UI is a web-based real-time chat application where users can join different chatrooms and send messages instantly. The system uses a REST API to manage chatrooms and WebSocket technology for real-time message delivery. The frontend is built with ReactJS, while the backend is powered by Node.js, NestJS, and TypeScript.







Welcome, **Person1**

CHANNELS

General
0 online

Random
2 online

[+ Create New Room](#)

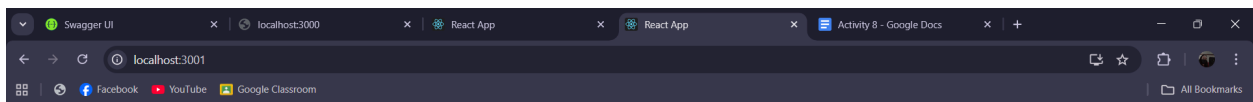
Live Chat

Person2
yo

Person1
uy

Type your message...

Send



Welcome, **Person1**

CHANNELS

General
0 online

Random
2 online

Hello World
0 online

New Room

Room name	Hello world2
Description (Optional)	
Icon (e.g., )	
Create	Cancel

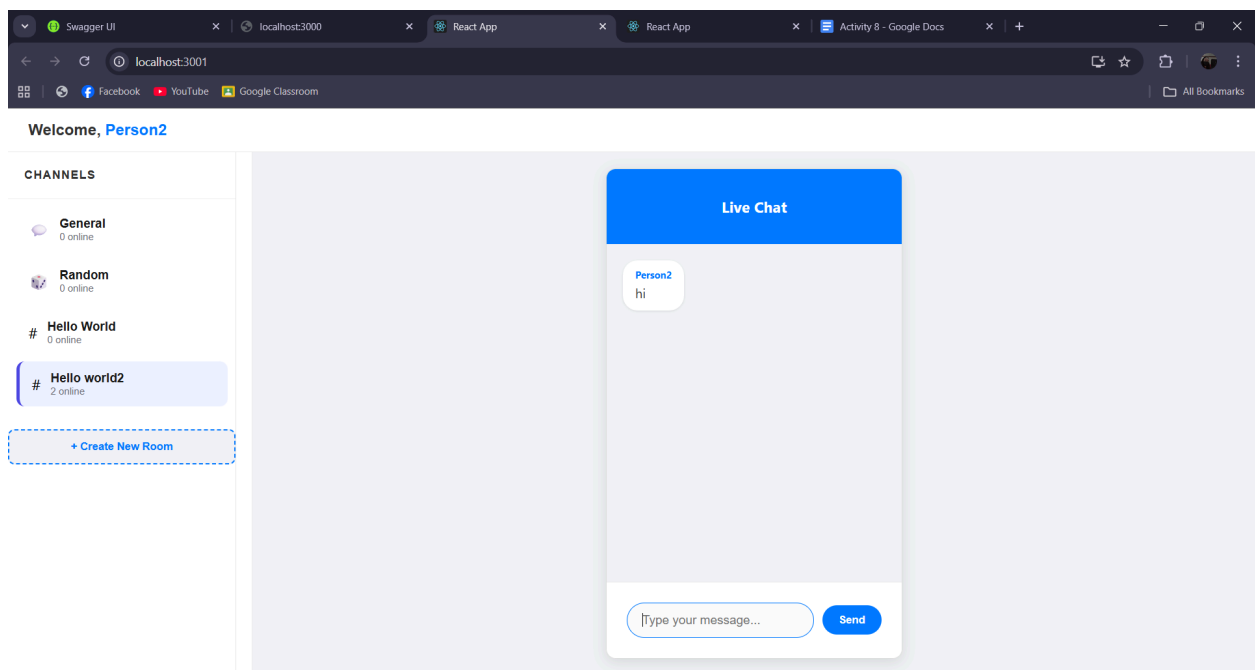
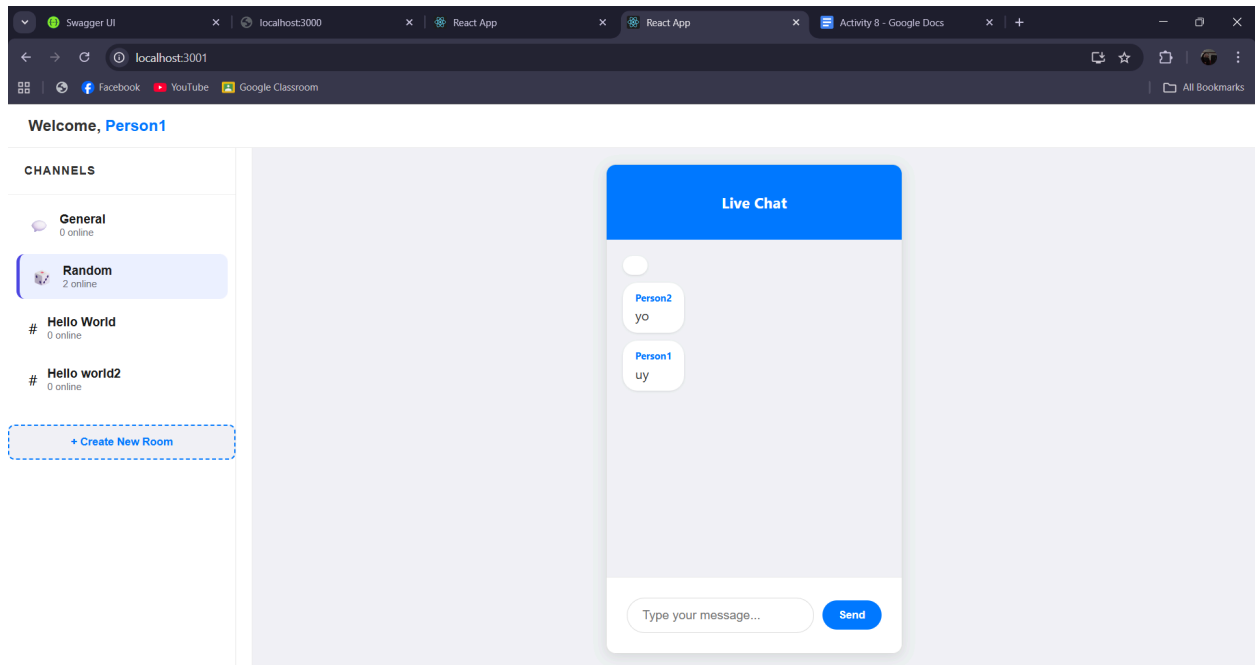
Live Chat

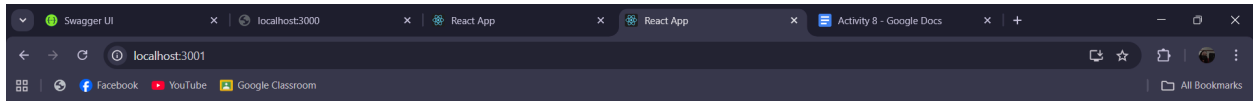
Person2
yo

Person1
uy

Type your message...

Send





Welcome, **Person1**

CHANNELS

 **General**
0 online

 **Random**
0 online

Hello World
0 online

Hello world2
2 online

[+ Create New Room](#)

Live Chat

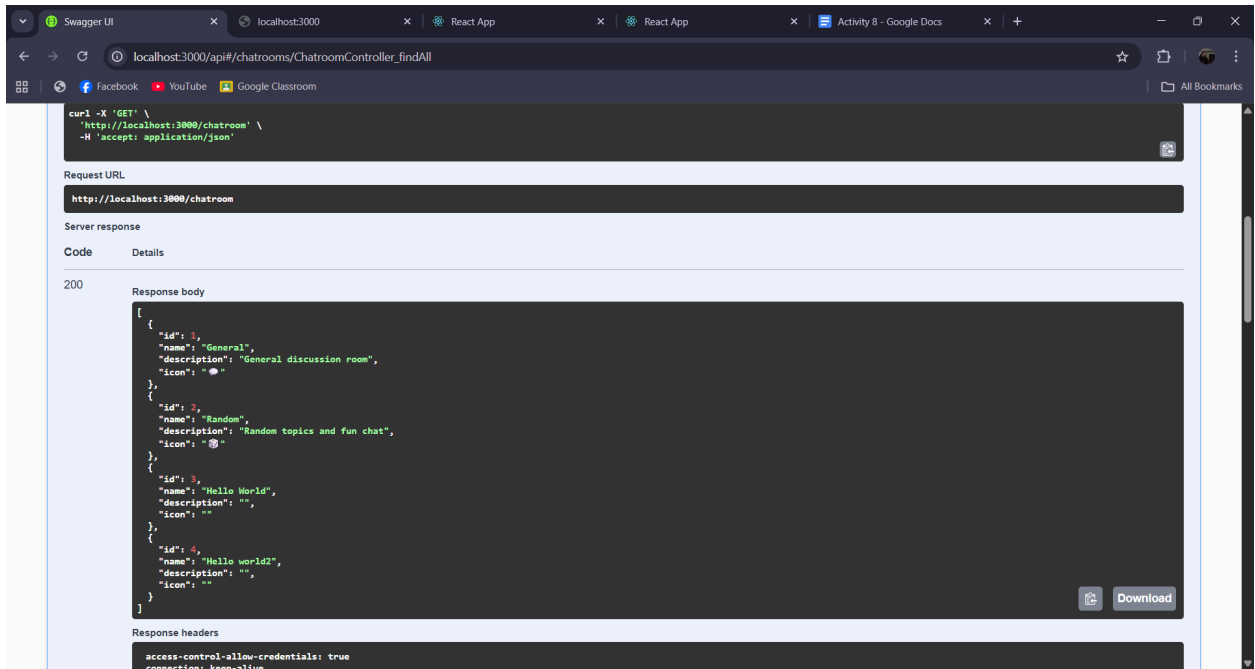
Person2
hi

Person1
q

Type your message...

Send

Chatroom CRUD:



Swagger UI interface showing the GET endpoint `localhost:3000/api#/chatrooms/ChatroomController_findAll`. The request URL is `http://localhost:3000/chatroom`. The server response is a 200 status code with a JSON array of chatroom objects in the response body. The response headers include `access-control-allow-credentials: true` and `connection: keep-alive`.

```
curl -X 'GET' \
  'http://localhost:3000/chatroom' \
  -H 'accept: application/json'
```

Request URL

http://localhost:3000/chatroom

Server response

Code Details

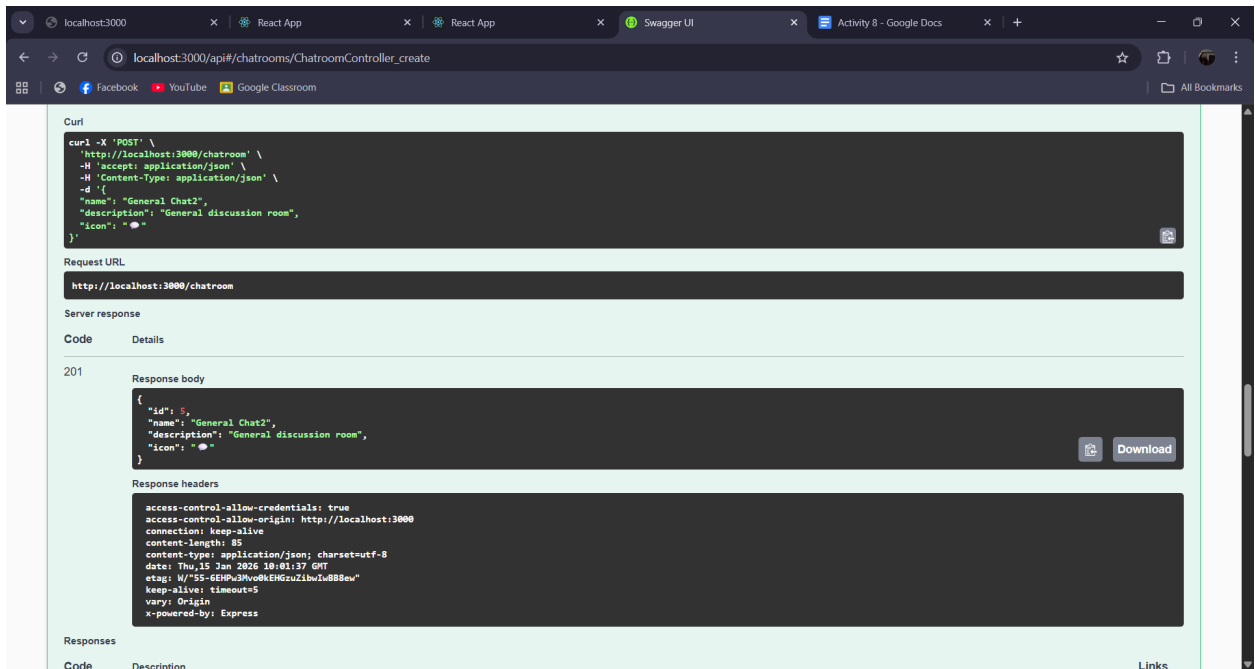
200

Response body

```
[
  {
    "id": 1,
    "name": "General",
    "description": "General discussion room",
    "icon": "👤"
  },
  {
    "id": 2,
    "name": "Random",
    "description": "Random topics and fun chat",
    "icon": "🎲"
  },
  {
    "id": 3,
    "name": "Hello World",
    "description": "",
    "icon": ""
  },
  {
    "id": 4,
    "name": "Hello world2",
    "description": "",
    "icon": ""
  }
]
```

Response headers

```
access-control-allow-credentials: true
connection: keep-alive
```



Swagger UI interface showing the POST endpoint `localhost:3000/api#/chatrooms/ChatroomController_create`. The request URL is `http://localhost:3000/chatroom`. The server response is a 201 status code with a JSON object in the response body. The response headers include `access-control-allow-credentials: true`, `access-control-allow-origin: http://localhost:3000`, `connection: keep-alive`, `content-length: 85`, `content-type: application/json; charset=utf-8`, `date: Thu, 15 Jan 2026 18:01:37 GMT`, `etag: W/\"151-68Pz3hVvohkEHGzuZlbuLd88ew\"`, `keep-alive: timeout=5`, `vary: Origin`, and `x-powered-by: Express`.

```
curl -X 'POST' \
  'http://localhost:3000/chatroom' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "General Chat2",
    "description": "General discussion room",
    "icon": "👤"
  }'
```

Request URL

http://localhost:3000/chatroom

Server response

Code Details

201

Response body

```
{
  "id": 5,
  "name": "General Chat2",
  "description": "General discussion room",
  "icon": "👤"
}
```

Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: http://localhost:3000
connection: keep-alive
content-length: 85
content-type: application/json; charset=utf-8
date: Thu, 15 Jan 2026 18:01:37 GMT
etag: W/\"151-68Pz3hVvohkEHGzuZlbuLd88ew\"
keep-alive: timeout=5
vary: Origin
x-powered-by: Express
```

Responses

Code	Description
------	-------------

Links

localhost:3000 x React App x React App x Swagger UI x Activity 8 - Google Docs x +

localhost:3000/api#/chatrooms/ChatroomController_findOne

Curl

```
curl -X 'GET' \
  'http://localhost:3000/chatroom/5' \
  -H 'accept: application/json'
```

Request URL

http://localhost:3000/chatroom/5

Server response

Code	Details
200	Response body <pre>{ "id": 5, "name": "General Chat2", "description": "General discussion room", "icon": "🗨️" }</pre> Response headers <pre>access-control-allow-credentials: true connection: keep-alive content-length: 85 content-type: application/json; charset=utf-8 date: Thu, 15 Jan 2026 18:02:12 GMT etag: W/"55-6EHPu3Mv00kEMGzuZibwTw888ev" keep-alive: timeout=5 vary: Origin x-powered-by: Express</pre>

Responses

Code	Description	Links
200	Chatroom found	No links

Media type

localhost:3000 x React App x React App x Swagger UI x Activity 8 - Google Docs x +

localhost:3000/api#/chatrooms/ChatroomController_update

Curl

```
curl -X 'PUT' \
  'http://localhost:3000/chatroom/5' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Updated Chat",
    "description": "Updated description",
    "icon": "🗨️"
  }'
```

Request URL

http://localhost:3000/chatroom/5

Server response

Code	Details
200	Response body <pre>{ "id": 5, "name": "Updated Chat", "description": "Updated description", "icon": "🗨️" }</pre> Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: http://localhost:3000 connection: keep-alive content-length: 88 content-type: application/json; charset=utf-8 date: Thu, 15 Jan 2026 18:02:44 GMT etag: W/"58-UW0DhZyK0svuJDh9fap1Q7NuRu" keep-alive: timeout=5 vary: Origin x-powered-by: Express</pre>

Responses

Code	Description	Links
------	-------------	-------

localhost:3000 React App React App Swagger UI Activity 8 - Google Docs

localhost:3000/api#/chatrooms/ChatroomController_remove

Responses

Curl

```
curl -X 'DELETE' \
  'http://localhost:3000/chatroom/5' \
  -H 'accept: */*'
```

Request URL

http://localhost:3000/chatroom/5

Server response

Code	Details
200	Response body <pre>{ "success": true }</pre> Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: http://localhost:3000 connection: keep-alive content-length: 16 content-type: application/json; charset=utf-8 date: Thu, 15 Jan 2026 18:03:13 GMT etag: W/"10-oV4hJcRVSEnc/wXsm4/Ped4A" keep-alive: timeout=5 vary: Origin x-powered-by: Express</pre>

Responses

Code	Description	Links
------	-------------	-------

Message CRUD:

localhost:3000 React App React App Swagger UI Activity 8 - Google Docs

localhost:3000/api#/messages/MessageController_findAll

Curl

```
curl -X 'GET' \
  'http://localhost:3000/message?chatroomId=1' \
  -H 'accept: application/json'
```

Request URL

http://localhost:3000/message?chatroomId=1

Server response

Code	Details
200	Response body <pre>[{ "id": 1, "chatroomId": 1, "username": "Person3", "message": "hi" }, { "id": 2, "chatroomId": 1, "username": "Person1", "message": "hello" }, { "id": 3, "chatroomId": 1, "username": "Person1", "message": "hello" }, { "id": 4, "chatroomId": 1, "username": "Person3", "message": "hi" }, { "id": 5, "chatroomId": 1, "username": "Person3", "message": "hi" }, { "id": 6, "chatroomId": 1, "username": "Person3", "message": "hi" }]</pre> Response headers <pre>access-control-allow-credentials: true connection: keep-alive</pre>

localhost:3000 x React App x React App x Swagger UI x Activity 8 - Google Docs x +

localhost:3000/api#/messages/MessageController_create

Facebook YouTube Google Classroom

cur1 -X 'POST' \

```
'http://localhost:3000/message' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "chatroomId": 1,
  "username": "John Doe",
  "message": "Hello, world!"
}'
```

Request URL

http://localhost:3000/message

Server response

Code	Details
201	Response body <pre>{ "id": 12, "chatroomId": 1, "username": "John Doe", "message": "Hello, world!" }</pre> Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: http://localhost:3000 connection: keep-alive content-length: 72 content-type: application/json; charset=utf-8 date: Thu, 15 Jan 2020 18:04:42 GMT etag: W/"48-hmW6H3UGMG7t8+whR10m8nqnJ0g" keep-alive: timeout=5 vary: Origin x-powered-by: Express</pre>

Responses

Code	Description	Links
201	Message created	No links

localhost:3000 x React App x React App x Swagger UI x Activity 8 - Google Docs x +

localhost:3000/api#/messages/MessageController_findOne

Facebook YouTube Google Classroom

Curl

```
curl -X 'GET' \
'http://localhost:3000/message/12' \
-H 'accept: application/json'
```

Request URL

http://localhost:3000/message/12

Server response

Code	Details
200	Response body <pre>{ "id": 12, "chatroomId": 1, "username": "John Doe", "message": "Hello, world!" }</pre> Response headers <pre>access-control-allow-credentials: true connection: keep-alive content-length: 72 content-type: application/json; charset=utf-8 date: Thu, 15 Jan 2020 18:05:23 GMT etag: W/"48-hmW6H3UGMG7t8+whR10m8nqnJ0g" keep-alive: timeout=5 vary: Origin x-powered-by: Express</pre>

Responses

Code	Description	Links
200	Message found	No links

localhost:3000

localhost:3000/api#/messages/MessageController_update

```
-H 'Content-type: application/json' \
-d '{
  "message": "Updated message text"
}'
```

Request URL

http://localhost:3000/message/12

Server response

Code	Details
200	Response body <pre>{ "id": 12, "chatroomId": 1, "username": "John Doe", "message": "Updated message text" }</pre> Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: http://localhost:3000 connection: keep-alive content-length: 79 content-type: application/json; charset=utf-8 date: Thu, 15 Jan 2026 18:05:54 GMT etag: W/"4f-uA12lg55685cMoIGhf1UJYxrxa" keep-alive: timeout=5 vary: Origin x-powered-by: Express</pre>

Responses

Code	Description	Links
200	Message updated	No links

Media type

application/json

localhost:3000

localhost:3000/api#/messages/MessageController_remove

Responses

Curl

```
curl -X 'DELETE' \
'http://localhost:3000/message/12' \
-H 'accept: */*'
```

Request URL

http://localhost:3000/message/12

Server response

Code	Details
200	Response body <pre>{ "success": true }</pre> Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: http://localhost:3000 connection: keep-alive content-length: 16 content-type: application/json; charset=utf-8 date: Thu, 15 Jan 2026 18:06:28 GMT etag: W/"18-oV4hJxRVSEhxc/uK3smA4/Pe4tA" keep-alive: timeout=5 vary: Origin x-powered-by: Express</pre>

Responses

Code	Description	Links
200	Message deleted successfully	No links

Instructions: How to Run the System

Start the Backend Server

Open a terminal

Go to the backend folder:

```
cd Activity-8/backend
```

Install dependencies:

```
npm install
```

Run the backend:

```
npm run start:dev
```

Make sure you see:

Application is running on <http://localhost:3000>

Start the Frontend (UI)

Open another terminal

Go to the frontend folder:

```
cd Activity-8/frontend
```

Install dependencies:

```
npm install
```

Run the frontend:

```
npm run start
```

Open the browser and go to:

<http://localhost:3001>

How to Use the Chatroom

1. Select or add a chatroom from the list

2. Open the same chatroom in another browser tab
3. Send a message
4. Messages will appear in real-time without refreshing

API Testing (Optional)

Use Postman to test the chatroom API:

- GET /chatrooms – view all chatrooms
- POST /chatrooms – add a new chatroom

Expected Output

- Chatrooms can be added via UI or API
- Messages are delivered in real-time
- No page refresh is needed